Assessing China’s Nuclear Capability

A Python-Based Open-Source Strategic Forces Study

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Open-source estimates only • No classified information • Portfolio / LinkedIn-ready project

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What the study asks

A strategic-force analysis framed for open-source research and public defense finance.

How has China’s nuclear capability changed — and what do public estimates suggest about its future role in deterrence?

600

Estimated Chinese nuclear warheads in 2025

1,000+

Public DoD assessment for 2030 trajectory

$12.5B

Estimated nuclear weapons spending in 2024

Presentation lens

Force structure: stockpile growth, global comparison, and delivery-system modernization
Forecasting: transparent low/base/high stockpile scenarios through 2035
Finance: public budget proxies and nuclear spending estimates with explicit limitations

Sources include SIPRI, FAS/Bulletin, CSIS ChinaPower, U.S. DoD reporting, ICAN estimates reported by Reuters/The Guardian, and official budget materials.

Method: a transparent open-source pipeline

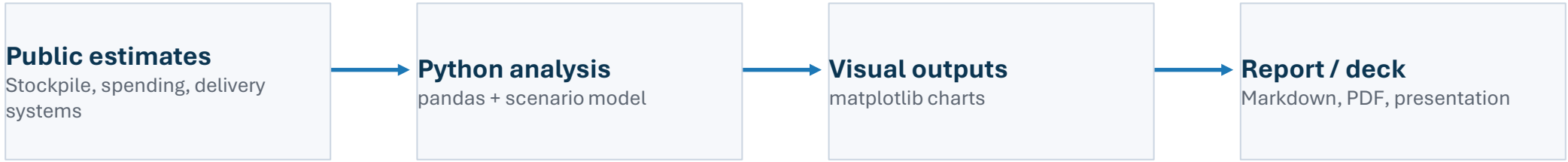
The project turns public estimates into charts, tables, scenarios, and a PDF report.

Input datasets

- China stockpile estimates
- Global nuclear force estimates
- Delivery-system public summary
- Defense and nuclear spending estimates
- Source confidence framework

Guardrails

Open-source estimates only. No classified information. No operational targeting, weapons design, or tactical employment guidance.

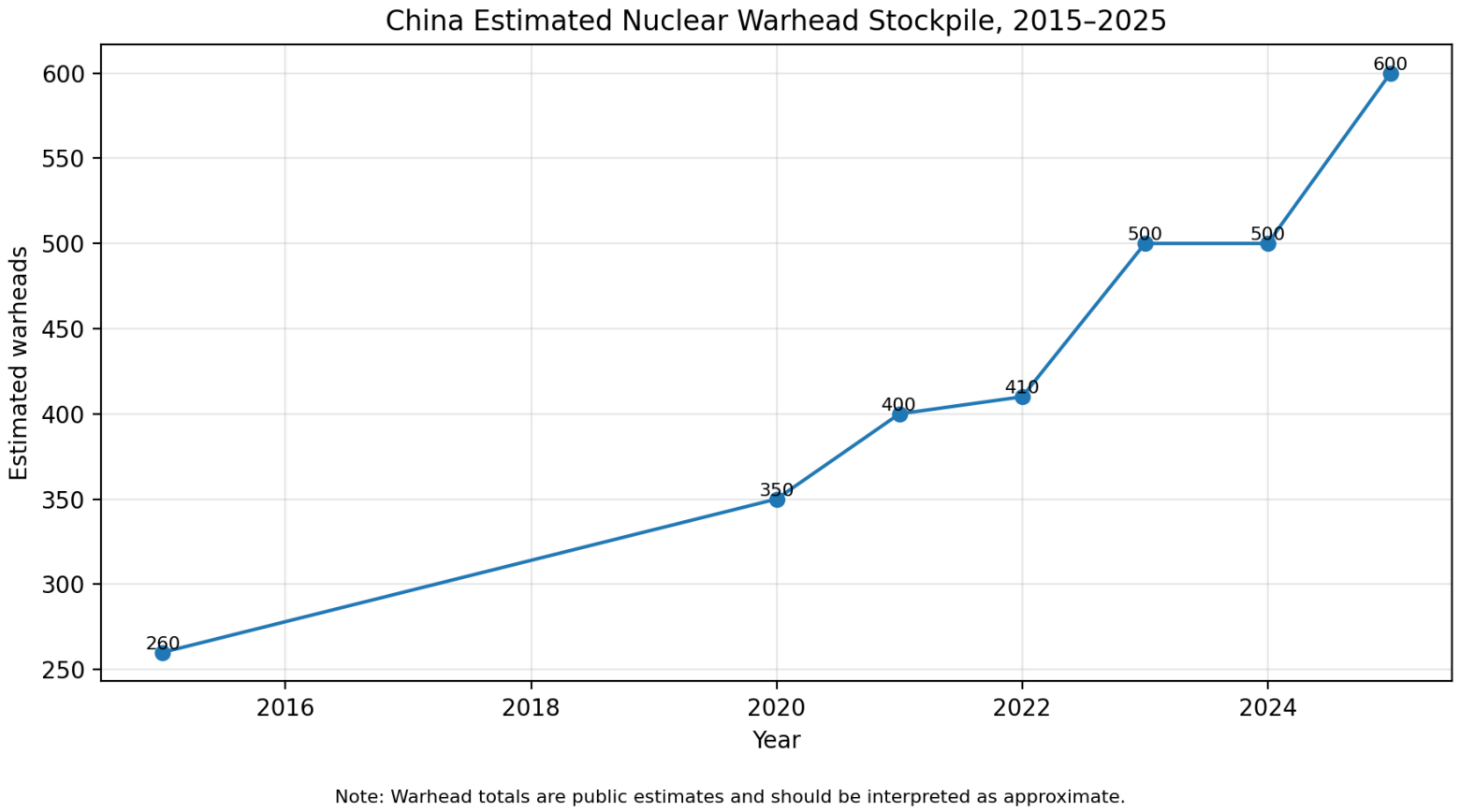


Why this matters: the project makes uncertainty visible instead of hiding it behind a single point estimate.

Research design: public-source datasets + transparent assumptions + reproducible Python chart generation.

Core finding: stockpile growth accelerated

Public estimates place China at roughly 600 nuclear warheads in 2025.



Note: Warhead totals are public estimates and should be interpreted as approximate.

600

Estimated warheads in 2025

+340

Approximate increase from 2015 baseline

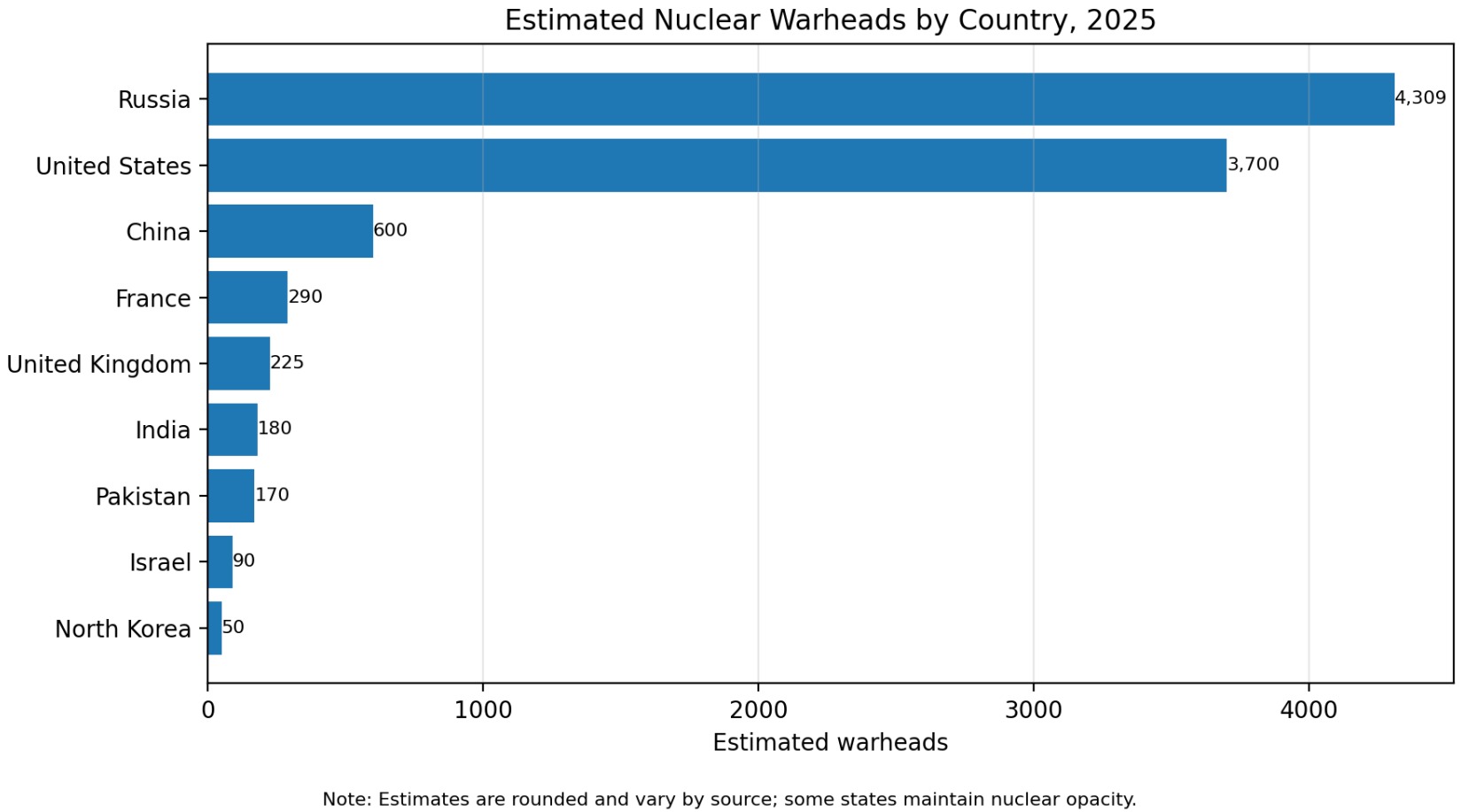
Interpretation

China remains far below the U.S. and Russia in total arsenal size, but the growth rate changes the strategic baseline.

Source framing: SIPRI Yearbook 2025; FAS/Bulletin Chinese Nuclear Weapons 2025; U.S. DoD reporting.

Global context: China is becoming the third major nuclear power

The U.S. and Russia remain much larger, but China now sits well above other nuclear-armed states.



Note: Estimates are rounded and vary by source; some states maintain nuclear opacity.

Key interpretation

Russia and the United States still dominate total warhead counts
China's public estimate is now materially above France, the U.K., India, Pakistan, Israel, and North Korea
The comparison is about scale, not operational readiness or posture

Note: Estimates are rounded and vary by source; some states maintain nuclear opacity.

Force structure: moving toward a more complete triad

Modernization spans land-based missiles, sea-based deterrence, and limited air-delivered roles.

Land-based

ICBM modernization

DF-31 family
DF-41
DF-26 dual-capable regional role

Sea-based

SSBN / SLBM deterrent

Type 094 SSBN
JL-3 SLBM reporting
Deployment details uncertain

Air-based

Limited bomber role

H-6N public reporting
Emerging/limited nuclear mission
Least mature leg

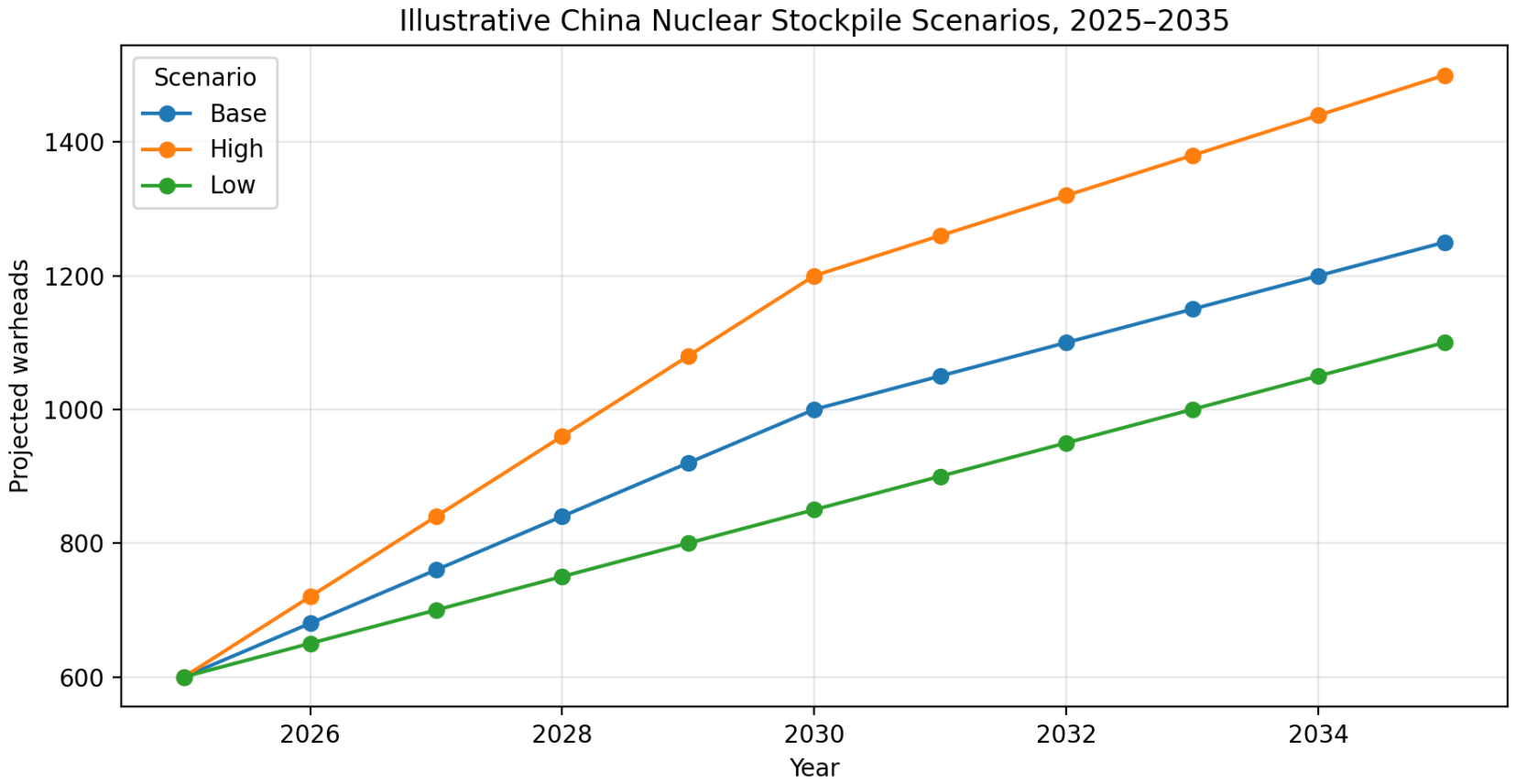
Strategic meaning

A more diversified force complicates deterrence calculations because survivability, regional reach, and escalation-management questions become more important.

Source framing: FAS/Bulletin and CSIS ChinaPower public reporting on delivery-system modernization.

Forecast scenarios: assumptions, not predictions

Base case reaches about 1,000 warheads by 2030; higher/lower paths show sensitivity.



Note: Scenarios are illustrative projections, not intelligence predictions; the base case broadly aligns with public DoD 2030 assessments.

1,000

Base scenario by 2030

1,500

High scenario by 2035

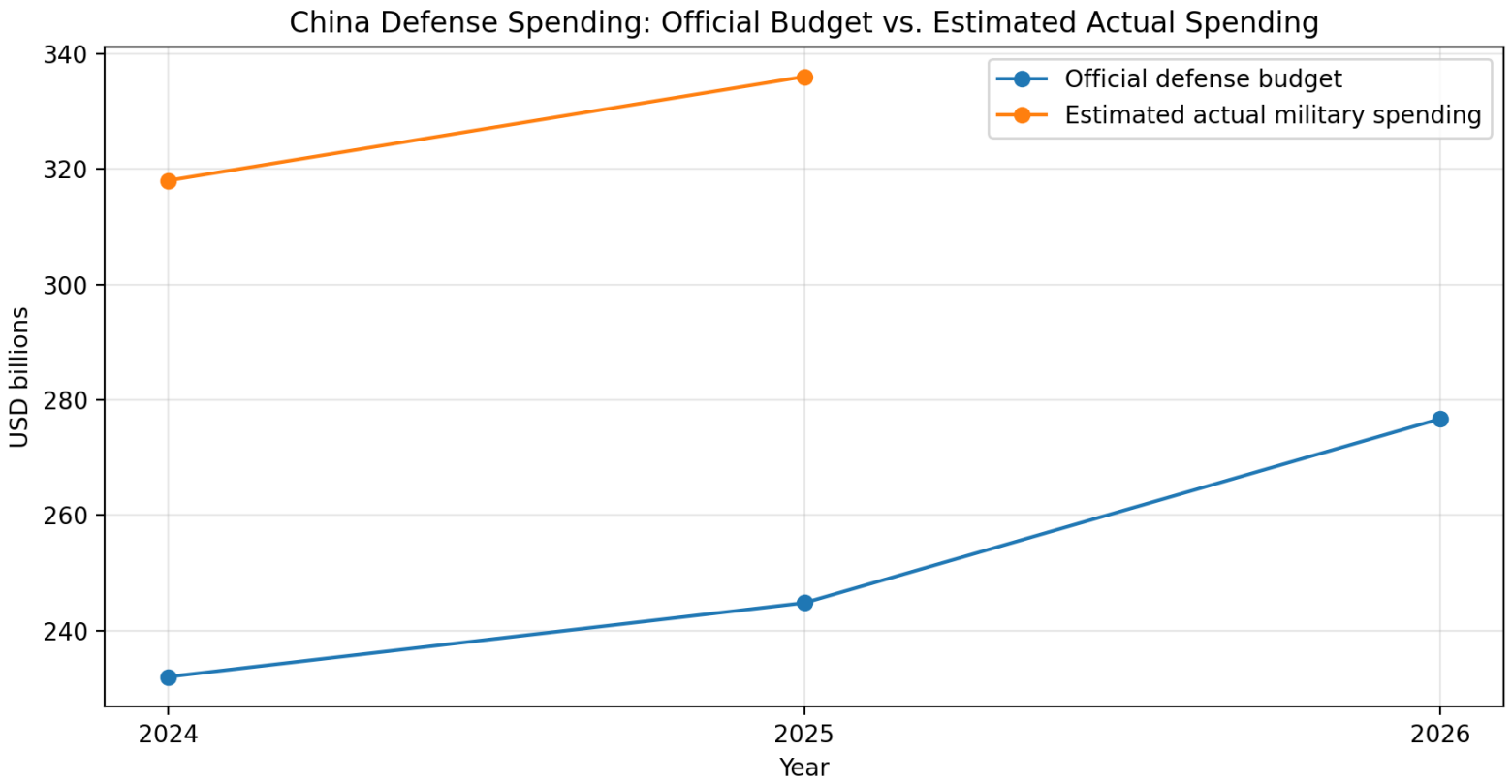
Why scenarios?

They make the assumptions explicit and avoid presenting uncertain growth paths as precise forecasts.

Note: The base case broadly aligns with public DoD 2030 assessments.

Budget context: official numbers are useful but incomplete

Defense finance can be used as a proxy, but it is not a direct nuclear-program measure.



Note: Official defense budgets may not capture total military-related spending; estimated actual spending uses outside public estimates.

$276.7B

2026 official defense budget estimate

$336B

2025 estimated actual military spending

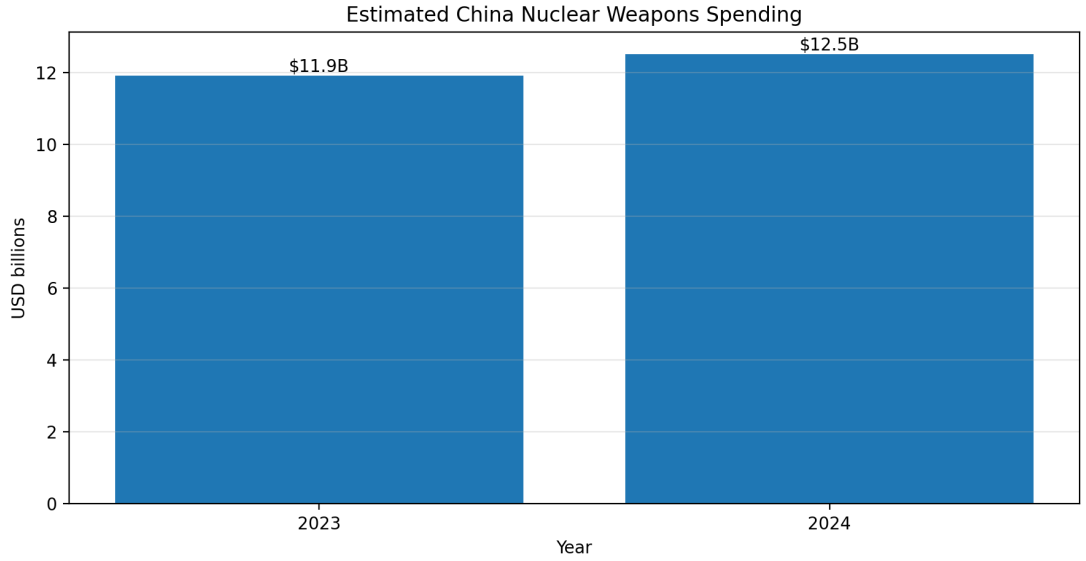
Interpretation

The gap between official and estimated actual spending is why the report treats budget numbers as proxies, not accounting totals.

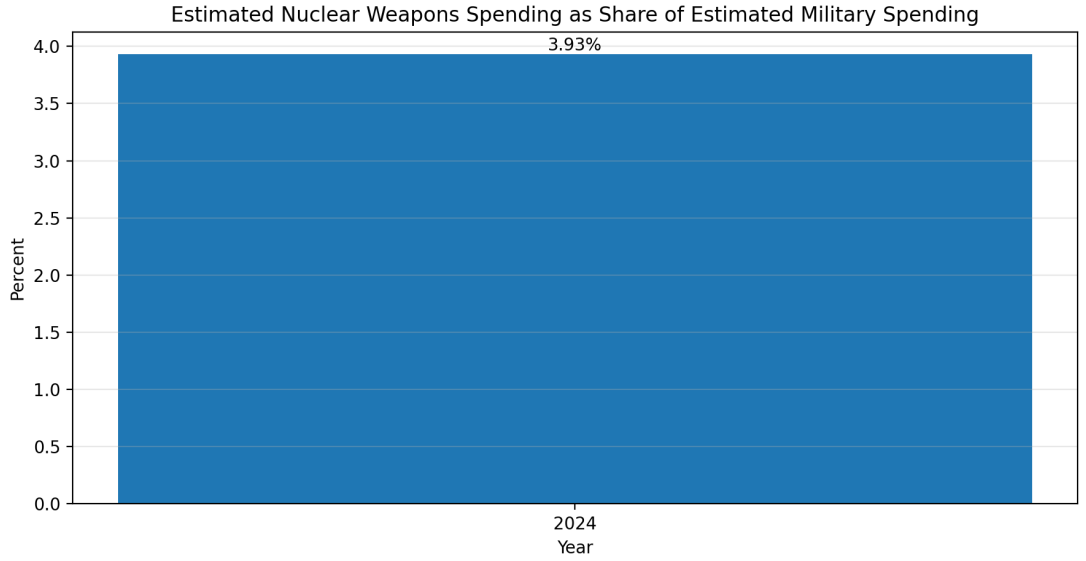
Source framing: CSIS ChinaPower, SIPRI military expenditure estimates, and Chinese official budget reporting.

Nuclear spending estimates: useful, but narrower data coverage

The public China-specific nuclear spending series currently covers only a limited number of years.



Note: China does not publish a separate nuclear weapons budget; figures are public estimates.



Note: Share uses estimated nuclear weapons spending divided by estimated actual military spending.

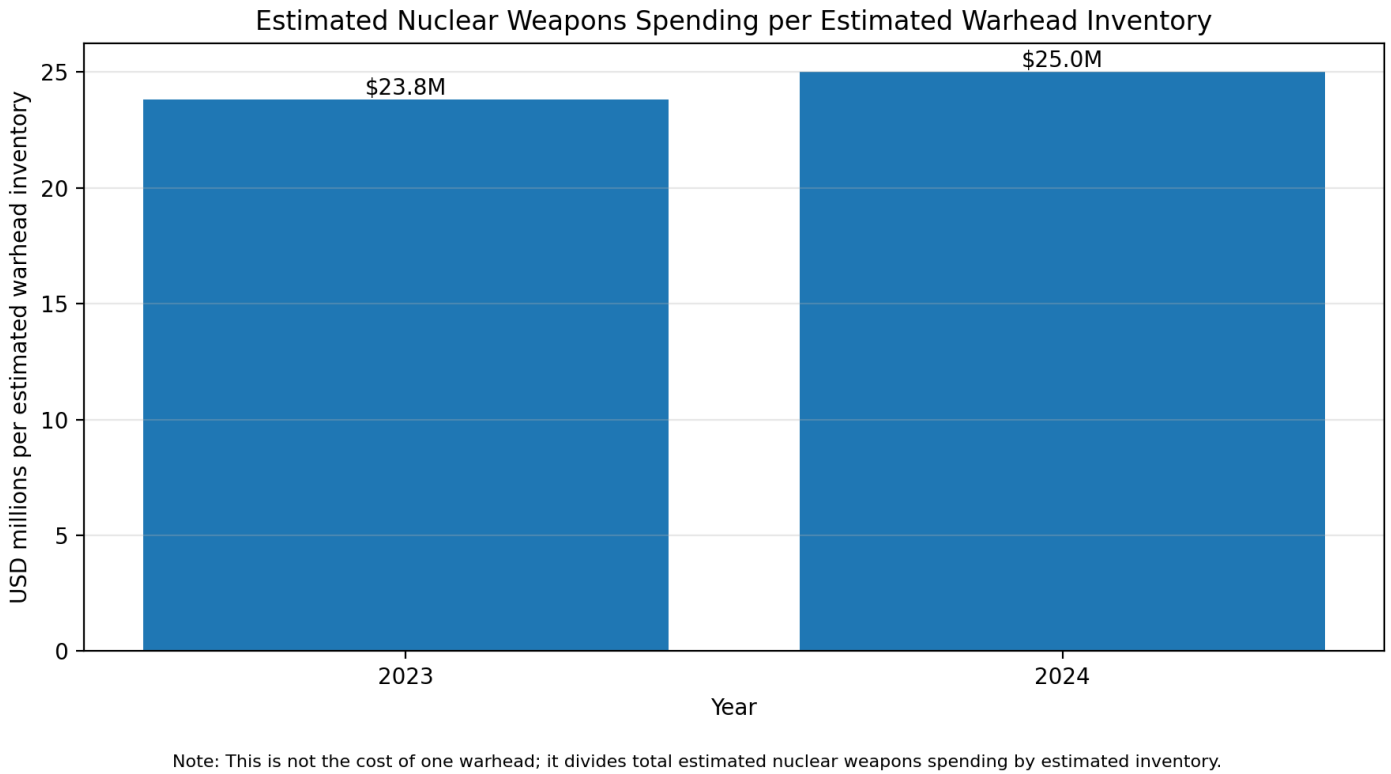
Reading the estimates

- 2024 estimated nuclear weapons spending: $12.5B
- Estimated share of actual military spending: 3.93%
- Directional indicator, not an official budget line

Source framing: ICAN nuclear weapons spending estimates reported by Reuters and The Guardian.

A derived metric — not the cost of a warhead

Spending per warhead inventory is a comparison metric, not a unit-cost estimate.



Note: This is not the cost of one warhead; it divides total estimated nuclear weapons spending by estimated inventory.

What it means

Divides estimated nuclear weapons spending by estimated warhead inventory
Useful for comparing public estimates across years
Does not isolate production cost, delivery systems, infrastructure, or personnel

Use carefully

This metric is included to show scale and transparency. It should not be used to infer procurement cost, deployment rate, or operational readiness.

Note: Nuclear spending can include warheads, delivery systems, submarines, infrastructure, command/control, research, production facilities, and personnel.

What makes the project defensible

The project separates findings from assumptions and documents uncertainty directly.

No classified data

All inputs are public estimates and government / research reporting.

Separate concepts

Official budgets, actual spending estimates, and nuclear spending estimates are not treated as the same thing.

Source confidence

Sources are ranked by institutional credibility, transparency, and usefulness for nuclear-force analysis.

Explicit limits

Stockpile counts, spending proxies, and readiness levels remain uncertain.

Source base includes SIPRI, FAS/Bulletin, CSIS, U.S. DoD, MOF budget reporting, ICAN estimates, BBVA, CRS.

Bottom line

China’s nuclear force is growing, diversifying, and becoming harder to evaluate with one number.

The project’s value is not only the finding - it is the reproducible workflow:
public data -> Python analysis -> charts -> report -> presentation.

7
charts embedded

12
source groups used

Open
reproducible methodology

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